

Passive Wi-Fi system to use 10,000 times less power than traditional Wi-Fi



A team from the engineering department at University of Washington, USA, has created a Wi-Fi system that uses 10,000 times less power than traditional Wi-Fi. Passive Wi-Fi, as the team calls it, is compatible with current smartphones and routers.

Traditional Wi-Fi radios consume a lot of power, mainly because of the analogue radio frequency (RF). In order to reduce power consumption of Wi-Fi radios in smartphones and digital devices, Passive Wi-Fi uses the digital baseband, while the analogue RF band is delegated to a single plugged-in device.

Essentially, Passive Wi-Fi helps remove the power-hungry analogue RF from the picture, which would make the smartphone last much longer.

The team has managed to achieve download speeds of up to 11Mbps, which falls in line with most LTE download speeds. However, the team is now working on increasing the throughput.

According to the team, this new technology will allow for new kinds of communications that was not possible before due to the power requirements of current Wi-Fi systems.

Super-tall robotic arm is made of helium balloons

Unlike regular robotic arms that you might find in a factory, this device, developed by Suzumori Endo Laboratory at Tokyo Institute of Technology and discovered by *IEEE Spectrum*, uses a series of helium-filled balloons to form its arm. The use of balloons means that the entire 20-metre-long structure weighs only 1.2 kilograms, which is light enough for simple pneumatic artificial muscles to be used to articulate its joints from the ground.

To further minimise the robot's weight, instead of using heavy pneumatic actuators that require hoses attached along the arm's length to pump hydraulic fluid, Giacometti Arm takes advantage of lightweight artificial muscles to move its individual segments.

However, there are some drawbacks to Giacometti Arm—presumably named after artist Alberto Giacometti, who is famous for his slender, spindly sculptures depicting the human form. Its lightweight nature means that it would be

easily buffeted by winds, for instance. Also, it cannot carry heavy stuff, either. And it is only as resilient as its balloons.



Giacometti Arm can easily squeeze into the trunk of a car (Image courtesy: screenshot taken from Youtube)